Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The sum of the rational terms in the expansion of $\left(\sqrt{2}+3^{1/5}\right)^{10}$ is equal to

(A) 40

(B) 41

(C) 42

(D) 0

Q.2

What are the values of k if the term independent of x in the expansion of $\left(\sqrt{x}+\frac{k}{x^2}\right)^{10}$ is 405 ?

(A) ± 3

(B) ± 6

(C) ± 5

(D) ± 4

Q.3

The greatest value of the term independent of x in the expansion of $\left(x \sin p+x^{-1} \cos p\right)^{10}, p \in \mathrm{R}$ is

(A) 2^5

(B) $\frac{10!}{2^5(5!)^2}$

(C) $\frac{10!}{(5!)^2}$

(D) None of these

Q.4

If the 7 th term in the binomial expansion of $\left(\frac{3}{\sqrt[3]{84}}+\sqrt{3} \ln x\right)^9, x>0$, is equal to 729, then x can be

(A) e^2

(B) e

(C) $\frac{e}{2}$

(D) $2e$

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Q.5

The number of term in the expansion of
$\left[(x+4y)^3(x-4y)^3\right]^2$ is

(A) 6

(B) 7

(C) 8

(D) 32

Q.6

The term independent of x in the expansion of
$\left[\left(t^{-1}-1\right)x+\left(t^{-1}+1\right)^{-1}x^{-1}\right]^8$ is

(A) $56\left(\frac{1-t}{1+t}\right)^3$

(B) $56\left(\frac{1+t}{1-t}\right)^3$

(C) $70\left(\frac{1-t}{1+t}\right)^4$

(D) $70\left(\frac{1+t}{1-t}\right)^4$

Q.7

If the middle term in the expansion of
$\left(\frac{1}{x}+x\sin x\right)^{10}$ equals to $7\frac{7}{8}$ then x is equal to; $(n\in I)$

(A) $2n\pi\pm\frac{\pi}{6}$

(B) $n\pi+\frac{\pi}{6}$

(C) $n\pi+(-1)^n\frac{\pi}{6}$

(D) $n\pi+(-1)^n\frac{5\pi}{6}$

Q.8

If the ratio of the 7th term from the beginning to
the 7th term from the end in $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n$ is $\frac{1}{6}$ them n equals to

(A) 10

(B) 9

(C) 8

(D) 12

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Q.9

The ninth term in the expansion of $\left\{3^{\log_3 \sqrt{25^{x-1}+7}}+3^{-1/8 \log_3\left(5^{x-1}+1\right)}\right\}^{10}$ is equal to 180 then x is

(A) a prime number

(B) an irrational number

(C) has non-zero fractional part

(D) None of these

Q.10

The approximate value of $(1.0002)^{3000}$ is

(A) 1.6

(B) 1.4

(C) 1.8

(D) 1.2

Q.11

The coefficient of x^{83} in $\left(1+x+x^2+x^3+x^4\right)^n(1-x)^{n+3}$, is $-{}^n\mathrm{C}_{2\lambda}$, then find the value of λ

(A) 12

(B) 10

(C) 9

(D) 8

Q.12

The interval in which x must lies so that the numerically greatest term in the expansion of $(1-x)^{21}$ has the greatest coefficient is, $(x>0)$

(A) $\left[\frac{5}{6}, \frac{6}{5}\right]$

(B) $\left(\frac{5}{6}, \frac{6}{5}\right)$

(C) $\left(\frac{4}{5}, \frac{5}{4}\right)$

(D) $\left[\frac{4}{5}, \frac{5}{4}\right]$

Q.13

The co-efficient of x^n in the expansion of $\frac{e^{7x}+e^x}{e^{3x}}$

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(A) $\frac{4^{n-1}+(-2)^n}{n!}$

(B) $\frac{4^{n-1}+2^n}{n!}$

(C) $\frac{4^n+(-2)^n}{n!}$

(D) $\frac{4^{n-1}+(-2)^{n-1}}{n!}$

Q.14

If $C_0, C_1, \ldots\ldots C_{15}$, are binomial coefficients in $(1+x)^{15}$
$\frac{C_1}{C_0}+2\frac{C_2}{C_1}+3\frac{C_3}{C_2}+\ldots..+15\frac{C_{15}}{C_{14}}=$

(A) 60

(B) 120

(C) 64

(D) 124

Q.15

If $x \neq 0$, then the sum of the series $1+\frac{x}{2!}+\frac{2x^2}{3!}+\frac{3x^3}{4!}+\ldots\ldots\ldots\infty$ is

(A) $\frac{e^x+1}{x}$

(B) $\frac{e^x(x-1)}{x}$

(C) $\frac{e^x(x-1)+1}{x}$

(D) $\frac{e^x(x-1)+1+x}{x}$

Answers & Solutions

Q.1 Answer:

41

Solution:

Here $T_{r+1}={}^{10}C_r(\sqrt{2})^{10-r}\left(3^{1/5}\right)^r$
where $r=0,1,2,\ldots.10$
We observe that in general term T_{r+1} **powers of**
2 and 3 are $\frac{1}{2}(10-r)$ **and** $\frac{1}{5}r$ **respectively and 0** $\leq r \leq 10$.
So, both these powers will be integers together only when $r=0$ **or 10**
$\therefore$ **Sum of required terms** $=T_1+T_{11}$
$={}^{10}C_0(\sqrt{2})^{10}+{}^{10}C_{10}\left(3^{1/5}\right)^{10}=32+9=41$

Q.2 Answer:

± 3

Solution:

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Given expansion is $\left(\sqrt{x}+\frac{k}{x^2}\right)^{10}$

$(r+1)_{\text{th}}$ **term,** $T_{\mathrm{r}+1}={}^{10}C_{\mathrm{r}}(\sqrt{x})^{10-r}\left(\frac{k}{x^2}\right)^r$

$\Rightarrow T_{r+1}={}^{10}C_r x^{5-r/2}\cdot(k)^r\cdot x^{-2r}$

$\therefore \mathrm{T_{r+1}}={}^{10}\mathrm{C_r x^{(10-5r)/2}(k)^r}$

Since, T_{r+1} **is independent of** x

$\therefore \frac{10-5r}{2}=0\Rightarrow r=2\therefore 405={}^{10}\mathrm{C}_2(\mathrm{k})^2$

$405=45\times \mathrm{k}^2\Rightarrow \mathrm{k}^2=9\Rightarrow \mathrm{k}=\pm 3$

Q.3 Answer:

$\frac{10!}{2^5(5!)^2}$

Solution:

' $\left(x\sin p+x^{-1}\cos p\right)^{10}$, **general term is**

$T_{r+1}={}^{10}C_r(x\sin p)^{10-\mathrm{r}}\left(x^{-1}\cos p\right)^r$

For the term independent of x **we have** $10-2r=0$

or $\quad r=5$

Hence, independent term is

${}^{10}C_5\sin^5 P\cos^5 P={}^{10}C_5\frac{\sin^5 2p}{32}$

which is greatest when $\sin 2p=1.$

Q.4 Answer:

e

Solution:

Let $r+1=7\Rightarrow r=6$

Given expansion is $\left(\frac{3}{\sqrt[3]{84}}+\sqrt{3}\ln x\right)^9, x>0$

We have $\mathrm{T}_{r+1}={}^{n}\mathrm{C}_r(x)^{n-r}a^r$ **for** $(x+a)^n$

$\therefore$ **According to the question**

$729={}^{9}\mathrm{C}_6\left(\frac{3}{\sqrt[3]{84}}\right)^3\cdot(\sqrt{3}\ln x)^6$

$\Rightarrow 3^6=84\times\frac{3^3}{84}\times 3^3\times(6\ln x)$

$\Rightarrow(\ln x)^6=1\Rightarrow(\ln x)^6=(\ln e)^6\Rightarrow x=\mathrm{e}$

Q.5 Answer:

7

Solution:

$\left[(x+4y)^3(x-4y)^3\right]^2=\left[\left\{x^2-(4y)^2\right\}\right]$

$=\left(x^2-16y^2\right)^6$

$\therefore$ **No. of terms in the expansion** $=7$

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Q.6 Answer:

$70\left(\frac{1-t}{1+t}\right)^4$

Solution:

$\left[\left(t^{-1}-1\right)x+\left(t^{-1}+1\right)^{-1}x^{-1}\right]^8$

$=\left[\left(\frac{1}{t}-1\right)x+\left(\frac{1}{t}+1\right)^{-1}\frac{1}{x}\right]^8$

Let T_{r+1} be the term independent of x, then

$\mathrm{T}_{r+1}={}^8\mathrm{C}_r\left(\frac{1}{t}-1\right)^{8-r}x^{8-r}\cdot\left(\frac{1}{t}+1\right)^{-r}\left(\frac{1}{x}\right)^r$

$={}^8\mathrm{C}_r\left(\frac{1}{t}-1\right)^{8-r}\cdot\left(\frac{1}{t}+1\right)^{-r}\cdot x^{8-2r}$

$\therefore 8-2r=0\Rightarrow r=4$

$\therefore$ **T_5 is the term independent of x and**

$\mathrm{T}_5={}^8C_4\left(\frac{1}{t}-1\right)^4\cdot\left(\frac{1}{t}+1\right)^{-4}$

$={}^8\mathrm{C}_4\left(\frac{1-t}{t}\right)^4\cdot\left(\frac{1+t}{t}\right)^{-4}$

$${}^8\mathrm{C}_4\left(\frac{1-t}{1+t}\right)^4=\frac{8.7\cdot6.5}{4\cdot3\cdot2\cdot1}\left(\frac{1-t}{1+t}\right)^4$$

$$=70\cdot\left(\frac{1-t}{1+t}\right)^4$$

Q.7 Answer:

$n\pi+(-1)^n\frac{\pi}{6}$

Solution:

Middle term in the expansion

$\left(\frac{10}{2}+1\right)^{\text{th}}$ **i.e.,** 6th **term.**

Thus $T_6=7\frac{7}{8}\Rightarrow{}^{10}C_5\frac{1}{x^5}\cdot x^5\sin^5x=\frac{63}{8}$

$\Rightarrow 252\cdot\sin^5x=\frac{63}{8}\Rightarrow\sin^5x=\frac{1}{32}$

$\Rightarrow\sin x=\frac{1}{2}$

$\therefore\quad x=n\pi+(-1)^n\frac{\pi}{6}$

Q.8 Answer:

9

Solution:

Given $\frac{T_7}{T_{n-7+2}}=\frac{1}{6}\Rightarrow\frac{T_7}{T_{n-s}}=\frac{1}{6}$

$$\Rightarrow\frac{{}^nC_6(\sqrt[3]{2})^{n-6}\left(\frac{1}{\sqrt[3]{3}}\right)^6}{{}^nC_{n-6}(\sqrt[3]{2})^6\left(\frac{1}{\sqrt[3]{3}}\right)^{n-6}}=\frac{1}{6}$$

$\Rightarrow 2^{\frac{n-12}{3}}\cdot3^{\frac{n-12}{3}}=\frac{1}{6}\Rightarrow6^{\frac{n-12}{3}}=6^{-1}$

$\therefore\quad\frac{n-12}{3}=-1\Rightarrow n=9$

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Q.9 Answer:

None of these

Solution:

We have,

$\left\{3^{\log_3 \sqrt{25^{x-1}+7}}+3^{-1/8 \log_3\left(5^{-1-1}+1\right)}\right\}^{10}$

$=\left[\sqrt{25^{x-1}+7}+\left(5^{x-1}+1\right)^{-1/8}\right]^{10}$

(**since** $a^{\log_a N}=N$)

Here, $T_9=180$

$\Rightarrow {}^{10}C_8\left\{\sqrt{25^{x-1}+7}\right\}^{10-8}\left\{\left(5^{x-1}+1\right)^{-1/8}\right\}^8=180$

$\Rightarrow {}^{10}C_8\left(25^{x-1}+7\right)\left(5^{x-1}+1\right)^{-1}=180$

$\Rightarrow 45\frac{\left(25^{x-1}+7\right)}{5^{x-1}+1}=180 \Rightarrow \frac{25^{x-1}+7}{5^{x-1}+1}=4$

$\Rightarrow \frac{y^2+7}{y+1}=4$, **where** $y=5^{x-1} \Rightarrow y^2-4y+3=0$

$\Rightarrow y=3,1 \Rightarrow 5^{x-1}=3$ **or** $5^{x-1}=1$

$\Rightarrow 5^x=15$ **or** $5^x=5 \Rightarrow x=\log_5 15$ **or** $x=1$

Q.10 Answer:

1.6

Solution:

$(1.0002)^{3000}=(1+0.0002)^{3000}$

$=1+(3000)(0.0002)+\frac{(3000)(2999)}{1.2}(0.0002)^2+$

$\frac{(3000)(2999)(2998)}{1.2.3}(0.0002)^3+\ldots$

We want to get answer correct to only one decimal places and as such we have left further expansion.

$=1+(3000)(0.0002)=1.6$

Q.11 Answer:

8

Solution:

We have, $\left(1+x+x^2+x^3+x^4\right)^n(x-1)^{n+3}$

$=\frac{\left(x^5-1\right)^n}{(1-x)^n}\cdot(x-1)^{n+3}=\left(x^5-1\right)^n(x-1)^3$

$=\left(+x^3-3x^2+3x-1\right)\sum_{r=0}^n {}^nC_r(-1)^r x^{5r}$

$=+\sum_{r=0}^n {}^nC_r(-1)^r x^{5r+3}+3\sum_{r=0}^n {}^nC_r(-1)^r x^{5r+2}$

$-3\sum_{r=0}^n {}^nC_r(-1)^r x^{5r+1}+3\sum_{r=0}^n {}^nC_r(-1)^r x^{5r}$

For term containing x^{83}**, we have** $5r+3=83 \Rightarrow r=16$ **whereas** $5r+2=83, 5r+1=83$ **and**

$5r=83$ **give no integral value of** r**. Hence, there is only one term containing** x^{83} **whose coefficient**

$=-{}^nC_{16}=-{}^nC_{2\lambda} \quad \therefore 2\lambda=16 \Rightarrow \lambda=8$

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Q.12 Answer:

$\left(\frac{5}{6}, \frac{6}{5}\right)$

Solution:

If n is odd, then numerically greatest coefficient in the expansion of $(1-x)^n$ is $\frac{{}^{n}C_{n-1}}{2}$ or $\frac{{}^{n}C_{n+1}}{2}$.

Therefore in $(1-x)^{21}$, the numerically greatest coefficient is ${}^{21}C_{10}$ or ${}^{21}C_{11}$. So, the numerically greatest term

$= {}^{21}C_{11}x^{11}$ **or** ${}^{21}C_{10}x^{10}$

So, $\left|{}^{21}C_{11}x^{11}\right| > \left|{}^{21}C_{12}x^{12}\right|$ **and**

${}^{21}C_{10}x^{10}| > |{}^{21}C_{9} \cdot x^{9}|$

$\Rightarrow \frac{21!}{10!11!} > \frac{21!}{9!12!}x$ **and** $\frac{21!}{11!10!}x > \frac{21!}{9!12!}$

$(\because x > 0)$

$\Rightarrow x < \frac{6}{5}$ **and** $x > \frac{5}{6} \Rightarrow x \in \left(\frac{5}{6}, \frac{6}{5}\right)$

Q.13 Answer:

$\frac{4^n+(-2)^n}{n!}$

Solution:

$\frac{e^{7x}+e^{x}}{e^{3x}} = e^{4x} + e^{-2x}$

$= \left[1 + 4x + \frac{(4x)^2}{2!} + \ldots\cdots\right] + \left[1 + (-2x) + \frac{(-2x)^2}{2!} + \ldots\cdots\right]$

$\therefore$ **coeff. of** $x^n = \frac{4^n}{n!} + \frac{(-2)^n}{n!}$

Q.14 Answer:

120

Solution:

General term of the given series is

$r\frac{{}^{n}C_{r}}{{}^{n}C_{r-1}} = n + 1 - r$

By taking summation over n, we get

$\sum_1^{15} r\frac{{}^{n}C_{r}}{{}^{n}C_{r-1}} = \sum_{n=1}^{15}(n+1-r) = \sum_1^{15}(16-r)$

$= 16 \times 15 - \frac{1}{2} \cdot 15 \times 16$

By using sum of n natural numbers $= \frac{n(n+1)}{2}$

$= 240 - 120 = 120$

Q.15 Answer:

$\frac{e^{x}(x-1)+1+x}{x}$

Solution:

The general term of the series

$\frac{x}{2!} + \frac{2x^2}{3!} + \frac{3x^3}{4!} + \ldots\ldots\infty$ **is**

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$T_n = \frac{nx^n}{(n+1)!}, n = 1, 2, \ldots\ldots\ldots\infty$

$= \frac{n+1-1}{(n+1)!}x^n = \frac{x^n}{n!} - \frac{1}{x}\frac{x^{n+1}}{(n+1)!}$

$1 + \frac{x}{2!} + \frac{2x^2}{3!} + \frac{3x^3}{4!} + \ldots\ldots$

$= 1 + \sum_{n=1}^{\infty}\frac{x^n}{n!} - \frac{1}{x}\sum_{n=1}^{\infty}\frac{x^{n+1}}{(n+1)!}$

$1 + (e^x - 1) - \frac{1}{x}(e^x - 1 - x)$

$\frac{xe^x - e^x + 1 + x}{x} = \frac{(x-1)e^x + (1+x)}{x}$

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